Irritable Bowel Syndrome (IBS)

Karen Pendergrass, Kimberly Eyer

Microbiome Signatures, Microbiome Medicine

Abstract. Discover IBS microbiome signatures, severity links, pediatric findings, and metal imbalances like copper–zinc ratios.

Published 2024-10-18 · https://microbiomemedicine.com/conditions/irritable-bowel-syndrome-ibs · A Microbiome Signatures condition monograph

Overview

Irritable Bowel Syndrome (IBS) is a common gastrointestinal disorder characterized by symptoms such as abdominal pain, bloating, and altered bowel habits. Recent research has focused on the gut microbiota's role in IBS, aiming to identify specific microbial signatures associated with the condition.

Key Insights

Altered Microbial Composition in IBS Patients:

Increased levels of Enterobacteriaceae, Lactobacillaceae, and Bacteroides; decreased levels of uncultured Clostridiales I, Faecalibacterium (including Faecalibacterium prausnitzii), and Bifidobacterium in IBS patients compared to healthy controls. [x] [x] [x] Increased Firmicutes, decreased Bacteroidetes, and a higher Firmicutes: Bacteroidetes ratio in IBS patients [x] and higher levels of Escherichia coli and Enterobacter, and lower levels of Lactobacillus and Bifidobacterium in IBS patients [4][x].

Pediatric Irritable Bowel

Increased abundance of γ-proteobacteria, particularly Haemophilus parainfluenzae, in pediatric IBS patients, a novel Ruminococcus-like microbe associated with IBS, and Increased abundance of several bacterial taxa from the genus Alistipes correlated with a greater frequency of pain in pediatric IBS patients [5]

Microbial Signatures and IBS Severity:

A specific microbial signature consisting of 90 bacterial operational taxonomic units (OTUs) associated with severe IBS symptoms. IBS symptom severity negatively associated with microbial richness, exhaled CH4, presence of methanogens, and enterotypes enriched with Clostridiales or Prevotella species. [6]

Metal Homeostasis in IBS

High serum copper levels are associated with a lower risk of developing IBS, suggesting a protective role of copper in IBS [7]

Individuals with IBS have higher copper–zinc ratios, suggesting a potential copper–zinc imbalance that may contribute to the underlying pathophysiology of IBS. [8]

References

1. Gut Microbiota in Patients With Irritable Bowel Syndrome-A Systematic Review.. Pittayanon, R., Lau, J., Yuan, Y., Leontiadis, G., Tse, F., Surette, M., & Moayyedi, P.. (Gastroenterology. 2019)
2. Gut Microbial Dysbiosis in the Irritable Bowel Syndrome: A Systematic Review and Meta-Analysis of Case-Control Studies.. Wang, L., Alammar, N., Singh, R., Nanavati, J., Song, Y., Chaudhary, R., & Mullin, G.. (Journal of the Academy of Nutrition and Dietetics. (2020))
3. Alterations of Gut Microbiota in Patients With Irritable Bowel Syndrome Based on 16S rRNA-Targeted Sequencing: A Systematic Review.. Duan, R., Zhu, S., Wang, B., & Duan, L.. (Gastroenterology. (2019))
4. Gut Microbial Dysbiosis in the Irritable Bowel Syndrome: A Systematic Review and Meta-Analysis of Case-Control Studies.. Wang, L., Alammar, N., Singh, R., Nanavati, J., Song, Y., Chaudhary, R., & Mullin, G.. (Journal of the Academy of Nutrition and Dietetics. (2020))
5. Gastrointestinal microbiome signatures of pediatric patients with irritable bowel syndrome.. Saulnier, D., Riehle, K., Mistretta, T., Diaz, M., Mandal, D., Raza, S., Weidler, E., Qin, X., Coarfa, C., Milosavljevic, A., Petrosino, J., Highlander, S., Gibbs, R., Lynch, S., Shulman, R., & Versalovic, J.. (Gastroenterology, 2011.)
6. dentification of an Intestinal Microbiota Signature Associated With Severity of Irritable Bowel Syndrome.. Tap, J., Derrien, M., Törnblom, H., Brazeilles, R., Cools-Portier, S., Doré, J., Störsrud, S., Nevé, B., Öhman, L., & Simrén, M.. (Gastroenterology, 2017.)
7. The Role of Copper and Zinc in Irritable Bowel Syndrome: A Mendelian Randomization Study.. Hujoel, I., & Hujoel, M.. (American journal of epidemiology. (2021))
8. Nutritional status in irritable bowel syndrome: A North American population-based study.. Hujoel, I.. (JGH Open: An Open Access Journal of Gastroenterology and Hepatology. 2020)

Generated from microbiomemedicine.com/conditions/irritable-bowel-syndrome-ibs. © Microbiome Signatures.